Data Changes Everything KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

1. How did you learn?

I learned to recognize Answers vary. from **rules** / **examples** (circle one).

2. ImageNet.

14 million
pictures

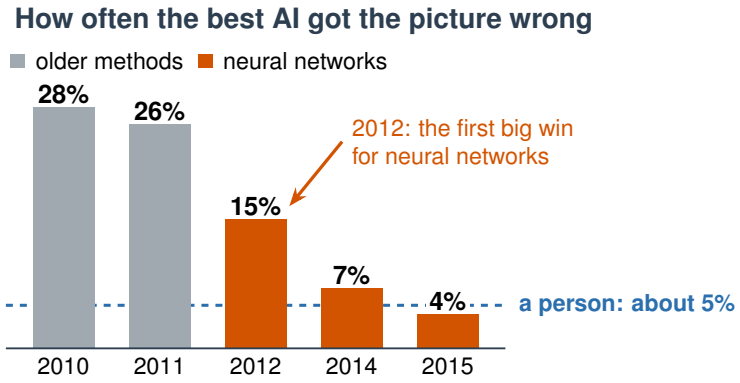
20,000+
kinds of things

49,000
people in 167 countries labeled them

ImageNet had about 14 million pictures of more than 20,000 kinds of things.

49,000 people in 167 countries looked at the pictures and gave each one a label.

3. The big win in 2012.



In 2012, a neural network won the contest. Its mistakes dropped from about 26% to 15%.

It was trained on GPUs: computer chips made for video games.

The new recipe for AI: 1. lots of data 2. neural networks 3. fast chips

4. Be the labelers. Label alone first. Then compare with a partner.

Round	**Pictures I said YES to** (numbers)	**Pictures we disagreed on**
chair	Answers vary.	
sandwich		
car		

Class vote. The hardest picture was _____. Yes: _____ No: _____

My rule for "sandwich": It is a sandwich if Answers vary. Example: food between two pieces of bread.

An AI learns what the labelers decided. It does not know what a sandwich *really* is.

5. Talk about it. 49,000 people labeled ImageNet. Most people do not know their names.

The people who labeled the data **should** / **should not** get credit, because Answers vary.

✂ *Cut out the cards. Add them to the class timeline.*

2009

Fei-Fei Li's team shares **ImageNet**: 14 million labeled pictures.

Draw a picture here.

2012

A **neural network** trained on video game chips wins the ImageNet contest.

Draw a picture here.

2016

AlphaGo beats Lee Sedol at Go, 4 games to 1.

Draw a picture here.